

Multiple Choice (10 marks)

1. Which of the following is NOT a source of atmospheric carbon?
 - (a) Respiration
 - (b) Photosynthesis
 - (c) Bacterial decomposition
 - (d) Combustion of fossil fuels
2. Which of the following depicts the correct sequence of membranes of the chloroplast, beginning with the innermost membrane and ending with the outermost membrane?
 - (a) Outer membrane, inner membrane, thylakoid membrane
 - (b) Thylakoid membrane, inner membrane, outer membrane
 - (c) Inner membrane, thylakoid membrane, outer membrane
 - (d) Thylakoid membrane, outer membrane, inner membrane
 - (e) Strama, outer membrane, inner membrane
3. Which statement about variation is true?
 - (a) All phenotypic variation is the result of genotypic variation.
 - (b) All genetic variation produces phenotypic variation.
 - (c) All nucleotide variability results in neutral variation.
 - (d) All new alleles are the result of nucleotide variability.
4. Most animal cells, regardless of species, are relatively small and about the same size. Relative to larger cells, why is this?
 - (a) Smaller cells are more resilient to external damage.
 - (b) Smaller cells have a larger number of organelles.
 - (c) Smaller cells require less energy to function.
 - (d) Smaller cells can multiply faster.
 - (e) Smaller cells fit together more tightly.
5. Short sequence by promoter that assists transcription by interacting with regulatory proteins.
 - (a) Promoter
 - (b) Inducer
 - (c) Repressor
 - (d) Operator
 - (e) Inhibitor

True / False (10 marks)

Answer True or False

6. True or False: Calcitonin acts antagonistically to parathormone by lowering blood calcium levels.
7. True or False: The probability of rolling a 5 on two successive rolls of a balanced die is $1/24$.
8. True or False: Flowers with strong sweet fragrances are primarily adapted to attract hummingbirds rather than bees.
9. True or False: Meiosis produces two haploid gametes after the completion of meiosis I.
10. True or False: Operant conditioning can be used to assess a dog's sensitivity to sounds of different frequencies through learned responses.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of removing a bacterial cell wall through lysozyme treatment. Can the bacteria survive, and how do bacterial and plant cell walls differ biochemically?
12. Discuss the role of plasma components, particularly inorganic salts, in influencing blood pressure. Additionally, analyze other solutes present in plasma and their potential effects.

End of Paper